JEE Mathematics: Quick Review with Grading and Practice Map - Limit (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (49 questions; a question that appears in two of these is counted once): High = 4 or more questions, Medium = 1 to 3 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Notion of limit

Question 01
Define limit.

Model Answer
If $f(x)$ comes as close as we like to a finite number l when x comes close to a (from both sides, $x \neq a$), then $\lim_{x\to a} f(x) = l$. The value $f(a)$ itself plays no part.

Ex: $\lim_{x\to 2} \dfrac{x^2-4}{x-2} = 4$ although the function is not defined at $x = 2$.

Grade - Low

Practice map, main concept: Sheet: C-II 1

Also used in: none

LHL, RHL and existence of limit

Question 02
What is LHL and RHL?

Model Answer
LHL = value approached by $f(x)$ as $x \to a^-$ $(x < a)$; RHL = value approached as $x \to a^+$ $(x > a)$.

Ex: $\lim_{x\to 2}[x]$: LHL $= 1$, RHL $= 2$.

Grade - Medium

Practice map, main concept: DPP: 3(5)

Also used in: Sheet: J-II 47; J-M 16; J-A 7; C-II 21 | DPP: 1(2, 12, 17) | Question Bank: 104

Question 03
Write all different ways in which LHL and RHL can be written.

Model Answer

$$\text{LHL} = \lim_{x\to a^-} f(x) = \lim_{h\to 0} f(a-h) = f(a^-) = f(a-0)$$

$$\text{RHL} = \lim_{x\to a^+} f(x) = \lim_{h\to 0} f(a+h) = f(a^+) = f(a+0)$$

Here $h \to 0^+$ (small positive).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 17, 20; J-II 47

Question 04
What is the condition for a limit to exist?

Model Answer
$\lim_{x\to a} f(x)$ exists $\iff$ LHL = RHL = a finite number. If either side is $\pm\infty$ or they differ, the limit does not exist (DNE).

Ex: $\operatorname{sgn} x$ at 0: LHL $= -1$, RHL $= 1$, so the limit does not exist.

Grade - Medium

Practice map, main concept: Sheet: C-I 25-26 | DPP: 1(12)

Also used in: Sheet: J-M 2; C-I 28

Question 05
What to keep in mind while computing limit if one side of the point is not in the domain?

Model Answer
Only the side lying in the domain is considered; the limit equals that one-sided limit (as used in JEE).

Ex: $\lim_{x\to 1} \sin^{-1} x = \lim_{x\to 1^-} \sin^{-1} x = \frac{\pi}{2}$ ($x > 1$ is not in the domain).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 4 | DPP: 2(17)

Indeterminate forms

Question 06
What do you mean by indeterminate form? State all 7 forms.

Model Answer
A form whose value cannot be decided by direct substitution; the expression must be simplified first.

$$\frac{0}{0},\ \frac{\infty}{\infty},\ \infty-\infty,\ 0\times\infty,\ 1^{\infty},\ 0^{0},\ \infty^{0}$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 07
How to check whether a form is an indeterminate form?

Model Answer
Put $x = a$ (or the limiting behaviour) in each part and see what each part tends to; it is indeterminate only if it matches one of the 7 forms with every 0, 1, ∞ only approached.

Ex: $\lim_{x\to\frac{\pi}{2}} \dfrac{\tan x}{\tan 3x}$ is $\frac{\infty}{\infty}$ (indeterminate); $\lim_{x\to 0^+} \dfrac{\sin x}{[x]}$ is not (the denominator is exactly 0).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 08

What is the condition on numbers in indeterminate forms?

Model Answer

The 0 and 1 in the forms are not exact: they are values being approached (0 means $\to 0$, 1 means $\to 1$).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 09

What is the value of the expression if a number gets its exact value in place of approaching value?

Model Answer

Then it is not indeterminate: exact $0\times$ anything $= 0$; $\frac{\text{exact } 0}{\text{anything}\neq 0} = 0$; $\frac{1}{\text{exact } 0}$ undefined; $(\text{exact } 1)^{\text{anything}} = 1$; $(\text{anything} \neq 0)^{\text{exact } 0} = 1$; $(\text{exact } 0)^{+ve} = 0$; $(\text{exact } 0)^0$ undefined.

Ex: $\lim_{x\to 0^+} [x] \cot x = 0$, since $[x] = 0$ exactly for $0 < x < 1$.

Grade - Low

Practice map, main concept: Sheet: J-I 21 | DPP: 5(1)

Also used in: DPP: 4(13-14)

Reading from graph

Question 10

How to find limit of a function if its graph is given?

Model Answer

Follow the curve towards $x = a$ from the left (LHL) and from the right (RHL) and read the heights; open/closed dots (the value $f(a)$) do not matter. If the curve goes up/down without bound, that side is $\pm\infty$ and the limit does not exist.

Grade - Low

Practice map, main concept: DPP: 1(2, 17)

Also used in: Sheet: C-I 23; C-II 29

Five fundamental theorem

Question 11

What will be limit of $f + g$, $f - g$, $f \cdot g$, f/g if limits of both f and g exist?

Model Answer

If $\lim f = l$, $\lim g = m$ (finite): $\lim(f \pm g) = l \pm m$, $\lim fg = lm$, $\lim \frac{f}{g} = \frac{l}{m}$ $(m \neq 0)$, $\lim kf = kl$.

Ex: $\lim_{x\to 0} \left(\frac{\sin x}{x} + e^x\right) = 1 + 1 = 2.$

Grade - Medium

Practice map, main concept: Sheet: C-I 6 | DPP: 1(1)

Also used in: Sheet: C-I 13 | Question Bank: 97

Question 12
What will be limit of $f+g$, $f-g$, $f\cdot g$, f/g if limits of both f and g do not exist?

Model Answer
Nothing can be said in general: the combination may or may not have a limit; compute it directly.
Ex: at $x=1$ neither $[x]$ nor $\{x\}$ has a limit, but $[x]+\{x\}=x\to 1$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 13
What will be limit of $f+g$, $f-g$, $f\cdot g$, f/g if limit of f exists and of g does not exist?

Model Answer
$f\pm g$: never exists. $f\cdot g$, f/g: may exist (possible only when $\lim f=0$ or g is bounded-type); check directly.
Ex: $\lim_{x\to 0} x\operatorname{sgn} x=0$ though $\operatorname{sgn} x$ has no limit at 0.

Grade - Low
Practice map, main concept: Sheet: C-I 13
Also used in: none

Strategies to calculate limits

Question 14
What are the basic steps to calculate limits?

Model Answer
Step 1: put $x=a$ and check the form. Determinate (e.g. $\frac{k}{\text{non-zero}}$, $\frac{k}{\infty}=0$) $\Rightarrow$ that is the answer.
Step 2: if indeterminate, simplify (factorise, rationalise, substitute, standard limits, expansion) until the form is removed. For [ ], { }, | |, sgn, ITF find LHL and RHL separately.
Ex: $\lim_{x\to 1}\dfrac{x^2+1}{x+2}=\dfrac{2}{3}$ directly.

Grade - Low
Practice map, main concept: Sheet: J-II 52, 54
Also used in: DPP: 1(15); 3(5); 4(21)

Question 15
What to keep in mind if we separate an expression while calculating limit?

Model Answer
Split $\lim(f\pm g)$ or $\lim fg$ into separate limits only if each part has a finite limit; otherwise you may create a false $\infty-\infty$ and a wrong answer.
Ex: in $\lim_{x\to 0}\dfrac{x-\sin x}{x^3}$ do not write $\lim\frac{x}{x^3}-\lim\frac{\sin x}{x^3}$; the answer is $\frac{1}{6}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 15
Also used in: Sheet: J-II 50; J-A 11 | DPP: 1(1)

Evaluating limit by factorization

Question 16

How to remove indeterminate forms by factorisation?

Model Answer

In $\frac{0}{0}$ of polynomials at $x = a$, $(x-a)$ is a factor of both; factorise, cancel the factor causing the $0/0$, then substitute.

Ex: $\lim_{x\to 3} \dfrac{x^2-2x-3}{x^2-4x+3} = \lim_{x\to 3} \dfrac{x+1}{x-1} = 2.$

Grade - **High**

Practice map, main concept: Sheet: J-I 13; J-II 1; J-M 20; J-A 9 | DPP: 1(4) | Question Bank: 99

Also used in: Sheet: J-I 1; J-II 32; J-M 10, 23; C-I 18 | DPP: 1(11, 15); 2(5) | Question Bank: 111

Question 17

What to keep in mind while factorizing expression while calculating limit?

Model Answer

Cancelling $(x-a)$ is allowed because $x \neq a$. Divide out $(x-a)$ fully (repeat if the form is still $\frac{0}{0}$); watch signs like $(a-x) = -(x-a)$ and $|x-a|$.

Ex: $\lim_{x\to 1} \dfrac{x^4-3x+2}{x^5-4x+3} = \lim_{x\to 1} \dfrac{x^3+x^2+x-2}{x^4+x^3+x^2+x-3} = 1.$

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 18

What to do in case of $\infty - \infty$ form?

Model Answer

Convert it to $\frac{0}{0}$ or $\frac{\infty}{\infty}$: take LCM of fractions, or rationalise roots, then proceed.

Ex: $\lim_{x\to 1} \left(\dfrac{1}{x-1} - \dfrac{3}{x^3-1} \right) = \lim_{x\to 1} \dfrac{x+2}{x^2+x+1} = 1.$

Grade - **Medium**

Practice map, main concept: Sheet: J-II 5; C-I 18; C-II 20 | DPP: 1(15)

Also used in: Sheet: J-II 30, 39; J-A 2 | DPP: 1(3); 2(4) | Question Bank: 96

Evaluating limit by rationalization

Question 19

How to remove indeterminate form by rationalisation?

Model Answer

Multiply numerator and denominator by the conjugate of the irrational part; the factor causing $\frac{0}{0}$ appears and cancels.

Ex: $\lim_{x\to 0} \dfrac{\sqrt{1+x}-\sqrt{1-x}}{x} = \lim_{x\to 0} \dfrac{2}{\sqrt{1+x}+\sqrt{1-x}} = 1.$

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 31 | DPP: 2(8)

Question 20

When and how to do square rationalisation?

Model Answer

When square roots give $\frac{0}{0}$ or $\infty - \infty$: use $a - b = \dfrac{a^2 - b^2}{a + b}$.

Ex: $\lim_{x\to\infty}\left(\sqrt{x^2+x} - x\right) = \lim_{x\to\infty} \dfrac{x}{\sqrt{x^2+x} + x} = \dfrac{1}{2}$.

Grade - High

Practice map, main concept: Sheet: J-II 13, 15, 39, 57; J-M 12; C-I 11 | DPP: 1(5, 9-10, 14); 2(5, 13) | Question Bank: 101

Also used in: Sheet: J-I 30; J-II 4, 10, 40; J-M 3, 7, 22; C-I 4; C-II 7, 32 | DPP: 1(3, 18); 2(12); 5(5) | Question Bank: 105, 110

Question 21

When and how to do cubic rationalisation?

Model Answer

When cube roots appear: use $a - b = \dfrac{a^3 - b^3}{a^2 + ab + b^2}$.

Ex: $\lim_{x\to 0} \dfrac{(1+x)^{1/3} - 1}{x} = \lim_{x\to 0} \dfrac{1}{(1+x)^{2/3} + (1+x)^{1/3} + 1} = \dfrac{1}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 22 | Question Bank: 110

Also used in: Sheet: J-II 6

Question 22

When can we put value of a factor while calculating limit?

Model Answer

Only when the factor is multiplied/divided with the rest and its own limit is finite and non-zero; never inside a sum or difference.

Ex: $\lim_{x\to 0} \dfrac{e^x(\sqrt{1+x} - 1)}{x} = e^0 \cdot \dfrac{1}{2} = \dfrac{1}{2}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 9 | DPP: 1(14)

Evaluating limit by substitution

Question 23

What to substitute if x approaches $-\infty$?

Model Answer

Put $x = -t$, $t \to +\infty$ (or $x = -\frac{1}{t}$, $t \to 0^+$). Remember $\sqrt{x^2} = |x| = -x$ for $x < 0$.

Ex: $\lim_{x\to-\infty} \dfrac{\sqrt{x^2+1}}{x} = \lim_{t\to\infty} \dfrac{\sqrt{t^2+1}}{-t} = -1$.

Grade - Low

Practice map, main concept: Sheet: J-II 20, 49 | DPP: 1(18)

Also used in: Sheet: J-II 30 | DPP: 1(9)

Question 24
What to substitute if x approaches ∞?

Model Answer
Put $x = \frac{1}{t}$, $t \to 0^+$; then standard limits / expansions at 0 can be used.

Ex: $\lim_{x\to\infty} x \sin\frac{1}{x} = \lim_{t\to 0^+} \dfrac{\sin t}{t} = 1$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 1; C-II 4
Also used in: Sheet: J-II 8, 29, 38; C-II 2, 16 | DPP: 2(7); 5(5, 9-10, 15, 17)

Question 25
What to substitute if x approaches a (non-zero finite)?

Model Answer
Put $x = a + h$, $h \to 0$ (or $x = a - h$ for one side), so the standard limits at 0 apply.
Ex: $\lim_{x\to\frac{\pi}{2}} \left(\frac{\pi}{2} - x\right) \tan x$: put $x = \frac{\pi}{2} - h$: $\lim_{h\to 0} h \cot h = 1$.

Grade - **Medium**
Practice map, main concept: Sheet: J-I 8; J-II 3, 6, 9; C-II 8 | DPP: 2(2, 4)
Also used in: Sheet: J-I 14; J-II 28; J-A 3 | DPP: 2(1, 9, 20); 3(2, 8); 4(7); 5(22) | Question Bank: 92

Infinity/infinity form

Question 26
How to solve questions of limit in case of ∞/∞ form?

Model Answer
Divide numerator and denominator by the highest power of x (or the largest exponential a^n) present; every other term then tends to 0.

Ex: $\lim_{x\to\infty} \dfrac{2x^2 + 3x}{5x^2 - 1} = \lim_{x\to\infty} \dfrac{2 + \frac{3}{x}}{5 - \frac{1}{x^2}} = \dfrac{2}{5}$.

Grade - **High**
Practice map, main concept: Sheet: J-I 3-4, 7, 9; J-II 48; J-M 1, 3, 6-7; C-I 20 | DPP: 1(3, 7-8, 13); 2(10, 14, 16) | Question Bank: 105
Also used in: Sheet: J-I 11, 15; J-II 15, 20, 49; J-M 19 | DPP: 1(10, 16) | Question Bank: 98

Question 27
What is the value of limit in questions of infinity/infinity form if highest power in numerator and denominator can be observed?

Model Answer

$$\lim_{x\to\infty} \frac{a_0 x^m + a_1 x^{m-1} + \dots}{b_0 x^n + b_1 x^{n-1} + \dots} = \begin{cases} \dfrac{a_0}{b_0}, & m = n \\ 0, & m < n \\ \pm\infty, & m > n \end{cases}$$

For $m > n$ the sign is that of $\frac{a_0}{b_0}$ (limit does not exist as a finite number).

Grade - **High**
Practice map, main concept: Sheet: J-M 19; C-II 12 | DPP: 1(6) | Question Bank: 98
Also used in: Sheet: J-II 34, 48; J-M 6; J-A 2 | DPP: 1(8); 5(23) | Question Bank: 95

Standard expansions

Question 28

Write expansion of $\sin x$, $\cos x$, $\tan x$, e^x, $\ln(1+x)$, $(1+x)^n$, $\tan^{-1} x$.

Model Answer

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots, \quad \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots, \quad \sin^{-1} x = x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \dots$$

$\ln(1+x)$, $\tan^{-1} x$, $(1+x)^n$ $(n \in Q)$ need $|x| < 1$; near $x \to 0$ this always holds.

Grade - High

Practice map, main concept: none

Also used in: Sheet: J-I 33, 35; J-M 11, 13, 17; C-I 29; C-II 33 | DPP: 5(9-10, 22) | Question Bank: 90-91, 94

Application of standard limits type 1

Question 29

Write result of standard limit type 1.

Model Answer

$$\lim_{x\to 0} \frac{(1+x)^n - 1}{x} = n, \qquad \lim_{x\to a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

Ex: $\lim_{x\to 0} \dfrac{(1+x)^{1/7} - 1}{x^4 + 4x} = \dfrac{1}{7} \cdot \dfrac{1}{4} = \dfrac{1}{28}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 30

What to replace if $(s^n - 1)$ appears in limit and what to keep in mind?

Model Answer

If $s \to 1$: replace $s^n - 1$ by $n(s-1)$ (equivalently $(1+t)^n - 1$ by nt when $t \to 0$). Allowed only when it is a factor of the numerator/denominator, not inside a sum.

Ex: $\lim_{x\to 1} \dfrac{x^{1/9} - 1}{x^2 - 3x + 2} = \lim_{x\to 1} \dfrac{\frac{1}{9}(x-1)}{(x-1)(x-2)} = -\dfrac{1}{9}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 56; J-A 1, 11; C-II 33 | Question Bank: 107

Also used in: Sheet: J-II 9; C-II 2, 5 | DPP: 2(2); 5(16)

Question 31

What is the value of limit $\lim_{x\to a}\dfrac{x^n-a^n}{x-a}$?

Model Answer

$$\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\quad(n\in Q,\ a>0)$$

Ex: $\lim_{x\to 2}\dfrac{x^5-32}{x-2}=5\cdot 2^4=80.$

Grade - Low

Practice map, main concept: Sheet: J-II 2; C-II 9 | DPP: 2(3)

Also used in: none

Trigonometric limits

Question 32

Write result of trigonometric limits.

Model Answer

$$\lim_{x\to 0}\frac{\sin x}{x}=\lim_{x\to 0}\frac{\tan x}{x}=\lim_{x\to 0}\frac{\sin^{-1}x}{x}=\lim_{x\to 0}\frac{\tan^{-1}x}{x}=1$$

x in radians. Ex: $\lim_{x\to 0}\dfrac{\sin 2x}{\sin 3x}=\dfrac{2}{3}.$

Grade - Medium

Practice map, main concept: Sheet: J-II 8; C-I 9 | Question Bank: 97

Also used in: Sheet: J-II 23, 45-46; C-I 1, 8; C-II 21, 24, 27 | DPP: 2(1)

Question 33

What to replace if $\sin(s)$, $\tan(s)$, $\sin^{-1}(s)$, $\tan^{-1}(s)$ appears in limit and what to keep in mind?

Model Answer

If $s\to 0$, replace each of them by s — only when it is a factor (product/quotient), never inside a sum or difference (there use expansions).

Ex: $\lim_{x\to 0}\dfrac{\sin x^2}{x\tan 2x}=\lim_{x\to 0}\dfrac{x^2}{x\cdot 2x}=\dfrac{1}{2}.$

Grade - High

Practice map, main concept: Sheet: J-M 9; J-A 4; C-II 3, 10, 13-14 | DPP: 2(6-7); 3(9, 11); 4(2) | Question Bank: 93

Also used in: Sheet: J-I 2-3, 5, 8, 10, 19; J-II 3, 53, 55-56; J-A 6, 12-13; C-I 12; C-II 18-19 | DPP: 2(16, 19); 3(10); 4(1); 5(16)

Question 34

What is the value of limit $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$?

Model Answer

$$\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}\quad(1-\cos s\approx\tfrac{s^2}{2}\text{ as }s\to 0)$$

Ex: $\lim_{x\to 0}\dfrac{1-\cos 4x}{x^2}=\dfrac{16}{2}=8.$

Grade - High

Practice map, main concept: Sheet: J-I 10, 14; J-II 4, 7; J-M 10, 23 | DPP: 2(1, 8-9, 11, 18) | Question Bank: 111

Also used in: Sheet: J-I 6, 8, 32; J-II 10, 14, 37-38, 46; J-M 4, 9, 12; J-A 5; C-I 3, 11; C-II 4-5, 8, 22, 30 | DPP: 2(4); 3(6, 10-11); 4(3, 21) | Question Bank: 101

Question 35

What is the value of limit of $x^p \sin\frac{1}{x}$, $x^p \cos\frac{1}{x}$ as $x \to 0$?

Model Answer

$\sin\frac{1}{x}$, $\cos\frac{1}{x}$ oscillate between -1 and 1 (no limit). So $\lim_{x\to 0} x^p \sin\frac{1}{x} = \lim_{x\to 0} x^p \cos\frac{1}{x} = 0$ if $p > 0$, and does not exist if $p \le 0$.

Ex: $\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0$ (bounded $\times \to 0$).

Grade - Medium

Practice map, main concept: Sheet: C-I 4 | DPP: 1(16); 4(8, 21)

Also used in: Sheet: J-A 7; C-II 23

Question 36

What to do if inverse trigonometric expression appears in limit question?

Model Answer

If its argument $\to 0$, use $\sin^{-1} s \approx s$, $\tan^{-1} s \approx s$. Otherwise put the ITF $= \theta$ (e.g. $\cos^{-1}(1-x) = \theta$), and keep the principal range and the side of approach in mind.

Ex: $\lim_{x\to 0^+} \dfrac{\cos^{-1}(1-x)}{\sqrt{x}}$: $1 - x = \cos\theta \Rightarrow x = 2\sin^2\frac{\theta}{2}$, limit $= \sqrt{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 19, 22-23; J-A 10; C-I 12, 16 | DPP: 2(15, 19-20); 4(19)

Also used in: Sheet: J-I 16-18; J-II 4, 27, 58; J-M 18; C-I 5, 10, 24, 26; C-II 6, 34 | DPP: 2(7); 3(3); 4(20)

Exponential and logarithmic limit

Question 37

Write result of exponential limit.

Model Answer

$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x\to 0} \frac{a^x - 1}{x} = \ln a \ (a > 0)$$

Ex: $\lim_{x\to 3} \dfrac{e^x - e^3}{x - 3} = e^3 \lim_{h\to 0} \dfrac{e^h - 1}{h} = e^3$.

Grade - Low

Practice map, main concept: Sheet: C-I 7 | DPP: 3(1-2)

Also used in: Sheet: C-I 15

Question 38

What to replace if $(e^s - 1)$, $(a^s - 1)$ appears in limit and what to keep in mind?

Model Answer

If $s \to 0$: $e^s - 1 \to s$, $a^s - 1 \to s \ln a$, only as a factor (product/quotient). Write $a^x - b^x = (a^x - 1) - (b^x - 1)$ first.

Ex: $\lim_{x\to 0} \dfrac{5^x - 3^x}{x} = \ln 5 - \ln 3 = \ln\dfrac{5}{3}$.

Grade - High

Practice map, main concept: Sheet: J-I 2; J-II 10-11; J-M 24; C-II 2 | DPP: 3(4, 6) | Question Bank: 112

Also used in: Sheet: J-II 9, 12; J-M 21; J-A 5, 8, 11; C-II 22 | DPP: 3(1, 11); 4(2) | Question Bank: 93, 100, 108

Question 39
Write result of logarithmic limit.

Model Answer

$$\lim_{x\to 0}\frac{\ln(1+x)}{x}=1, \qquad \lim_{x\to 0}\frac{\log_a(1+x)}{x}=\frac{1}{\ln a}$$

Ex: $\lim_{x\to 0}\dfrac{\ln(1+3x)}{x}=3.$

Grade - Low

Practice map, main concept: none

Also used in: Sheet: C-I 18

Question 40
What to replace if $\ln(1+s)$ appears in limit and what to keep in mind?

Model Answer
If $s\to 0$: $\ln(1+s)\to s$; equivalently if $S\to 1$: $\ln S\to S-1$. Only as a factor; inside sums use the expansion.

Ex: $\lim_{x\to 1}\dfrac{\ln x}{x-1}=1.$

Grade - Medium

Practice map, main concept: Sheet: J-I 1, 12 | DPP: 3(10)

Also used in: Sheet: J-I 2; J-II 1, 9, 58; J-M 9; C-I 21; C-II 4, 15 | DPP: 3(1, 9, 11) | Question Bank: 93

One raise to infinity form

Question 41
How to check whether indeterminate form is 1^∞ form?

Model Answer
In $f(x)^{g(x)}$ the base $f\to 1$ (not exactly 1) and the exponent $g\to\pm\infty$.
Ex: $\lim_{x\to 0}(\cos x)^{1/x^2}$ is 1^∞; $\lim_{x\to 0}(\operatorname{cosec} x)^{1/x}$ is not (base $\to\infty$).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 42
What to replace in case of 1^∞ form?

Model Answer

$$\lim f(x)^{g(x)}=e^{\lim g(x)\,(f(x)-1)} \qquad (f\to 1,\ g\to\infty)$$

Ex: $\lim_{x\to 0}(\cos x)^{1/x^2}=e^{\lim\frac{\cos x-1}{x^2}}=e^{-1/2}.$

Grade - High

Practice map, main concept: Sheet: J-I 5-6, 11; J-II 12, 14, 38, 40, 53; J-A 12; C-I 15, 21; C-II 5, 7, 22, 34 | DPP: 3(3, 7-8, 12-13); 5(3, 11, 16, 23) | Question Bank: 95

Also used in: Sheet: J-I 34; J-II 36; J-M 24; C-II 13, 16, 32-33 | DPP: 4(15-16); 5(8, 17) | Question Bank: 94, 112

Questions based on derivatives

Question 43

Write result of $f'(x)$, $f''(x)$, $f'''(x)$, $f''''(x)$ as limits.

Model Answer

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}, \quad f''(x) = \lim_{h\to 0} \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}$$

$$f'''(x) = \lim_{h\to 0} \frac{f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)}{h^3}$$

$$f''''(x) = \lim_{h\to 0} \frac{f(x+4h) - 4f(x+3h) + 6f(x+2h) - 4f(x+h) + f(x)}{h^4}$$

Coefficients follow $(h-1)^n$. Ex: $\lim_{h\to 0} \dfrac{\tan(x+h) - \tan x}{h} = \sec^2 x$.

Grade - Medium

Practice map, main concept: Sheet: J-I 30-31; J-II 25-26; J-A 14 | DPP: 4(3)

Also used in: none

Special type of functions

Question 44

How to compute limit of a function defined separately for rational and irrational numbers?

Model Answer

Near any point there are both rationals and irrationals, so the limit exists only at points where both definitions tend to the same value: solve $f_1(a) = f_2(a)$.

Ex: $f(x) = x^2$ $(x \in Q)$, 0 $(x \notin Q)$: limit exists only at $x = 0$ (value 0); at $x = 1$ it does not exist.

Grade - Low

Practice map, main concept: DPP: 4(4, 17)

Also used in: none

Problems based on special signs

Question 45

What to do if modulus appears in limit?

Model Answer

Find LHL and RHL separately: open $|g(x)|$ as g or $-g$ according to the sign of g on each side.

Ex: $\lim_{x\to 0} \dfrac{\sin x}{|x|}$: LHL $= -1$, RHL $= 1$, does not exist.

Grade - Medium

Practice map, main concept: Sheet: J-II 47; J-A 7; C-I 2 | DPP: 4(1)

Also used in: Sheet: J-II 19; J-M 14; J-A 14; C-I 24-26 | DPP: 4(11-12); 5(1) | Question Bank: 102

Question 46

What to do if $[\cdot]$ appears in limit question?

Model Answer

Find LHL and RHL. If the inside tends to an integer k, see whether it approaches from below ($[\ ] = k-1$) or above ($[\ ] = k$); if the inside tends to a non-integer, put its value.

Ex: $\lim_{x\to 2}[x^2]$: LHL $= 3$, RHL $= 4$; $\lim_{x\to 2}[x]^2$: LHL $= 1$, RHL $= 4$.

Grade - High

Practice map, main concept: Sheet: J-I 16-18, 20; J-II 17, 51; J-M 2, 16; C-I 24, 28; C-II 32 | DPP: 4(10, 13-14, 16, 20) | Question Bank: 104

Also used in: Sheet: J-I 21; J-II 19, 52, 56; J-M 5, 15; C-I 19, 25; C-II 25, 31 | DPP: 3(3); 4(6, 19) | Question Bank: 103

Question 47

What to do if $\{\cdot\}$ appears in limit question?

Model Answer

Write $\{x\} = x - [x]$ and find LHL and RHL; $\{\ \}$ is periodic with period 1 and $0 \le \{x\} < 1$.

Ex: $\lim_{x\to 0}\{x\}$: LHL $= 1$, RHL $= 0$, does not exist.

Grade - Low

Practice map, main concept: Sheet: J-I 19; C-I 17, 19 | DPP: 4(15)

Also used in: Sheet: J-II 45, 51; C-II 30 | DPP: 4(16, 20)

Question 48

What to replace $\{x\}$ when x approaches any integer?

Model Answer

$$x \to n^-:\ \{x\} = 1 - h \to 1; \qquad x \to n^+:\ \{x\} = h \to 0 \quad (n \in I)$$

Ex: $\lim_{x\to 2^-}\{x\} = 1$, $\lim_{x\to 2^+}\{x\} = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-M 18; C-II 6 | DPP: 4(6); 5(4)

Also used in: Sheet: J-I 19; J-II 51 | DPP: 4(2)

GIF on standard limits

Question 49

In standard limit questions, value of y approaches answer from which side?

Model Answer

From $\sin x < x < \tan x$ $(0 < |x| < \frac{\pi}{2})$: $\dfrac{\sin x}{x} \to 1^-$, $\dfrac{\tan x}{x} \to 1^+$, $\dfrac{\sin^{-1} x}{x} \to 1^+$, $\dfrac{\tan^{-1} x}{x} \to 1^-$ (reciprocals reverse the side); $\cos x \to 1^-$.

Ex: $\lim_{x\to 0}\left[\dfrac{\sin x}{x}\right] = 0$, $\lim_{x\to 0}\left[\dfrac{\tan x}{x}\right] = 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 22; J-II 18; J-M 4 | DPP: 4(7, 9, 11-12)

Also used in: Sheet: J-II 44; C-I 23, 27 | DPP: 5(19)

Sandwich theorem

Question 50

Write statement of Sandwich theorem.

Model Answer

If $g(x) \le f(x) \le h(x)$ near a and $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = l$, then $\lim_{x\to a} f(x) = l$ (a, l may be $\pm\infty$). If the two bounds have different limits, the theorem gives nothing.

Ex: $5 - 2x^2 \le f(x) \le 5 - x^2 \Rightarrow \lim_{x\to 0} f(x) = 5$.

Grade - Medium

Practice map, main concept: Sheet: J-I 28; J-M 8, 15 | Question Bank: 103, 106

Also used in: none

Question 51

How to solve sandwich theorem questions involving $[\cdot]$?

Model Answer

Use $x - 1 < [x] \le x$ for each term, multiply/add, then take limits of both bounds.

Ex: $\lim_{x\to 0^+} x\left[\frac{1}{x}\right] = 1$ (from $1 - x < x[\frac{1}{x}] \le 1$); $\lim_{n\to\infty} \dfrac{[x] + [2x] + \cdots + [nx]}{n^2} = \dfrac{x}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 23-24, 27, 29 | Question Bank: 109

Also used in: Sheet: J-I 25

Question 52

How to solve sandwich theorem questions involving series sum?

Model Answer

For n terms: $n \times$ (smallest term) $\le S_n \le n \times$ (largest term); if both bounds tend to the same value, that is the limit.

Ex: $\dfrac{n}{\sqrt{n^2+n}} \le \sum_{r=1}^{n} \dfrac{1}{\sqrt{n^2+r}} \le \dfrac{n}{\sqrt{n^2+1}}$, so the limit is 1.

Grade - Low

Practice map, main concept: Sheet: J-I 26; J-II 24 | DPP: 4(5, 18)

Also used in: none

Limits of composite function

Question 53

How to find limit of a composite of two functions without finding the composite?

Model Answer

For $\lim_{x\to a} f(g(x))$: find what $g(x)$ tends to and from which side ($g(a^-)$, $g(a^+)$), then take f at that value from that side.

Ex: $\lim_{x\to 0}[\cos x] = 0$ since $\cos x \to 1^-$.

Grade - Medium

Practice map, main concept: Sheet: J-II 21, 44-45; J-M 5; C-I 14, 22-23, 27; C-II 29-31 | DPP: 2(17); 5(2, 19-21)

Also used in: none

Question 54

How to find limit when numerator and denominator are expressed as expansion of x?

Model Answer

$$\lim_{x\to 0}\frac{a_0x^n+a_1x^{n+1}+\dots}{b_0x^m+b_1x^{m+1}+\dots}=\begin{cases}\dfrac{a_0}{b_0}, & n=m\\ 0, & n>m\\ \text{DNE}, & n<m\end{cases}$$

Compare the lowest powers. Ex: $\lim_{x\to 0}\dfrac{x^7+3x^3}{x^5-2x^3}=-\dfrac{3}{2}$.

Grade - High

Practice map, main concept: Sheet: J-I 33, 35; J-II 28-29, 42-43, 58; C-II 11, 16 | DPP: 5(7, 10, 14-15, 17) | Question Bank: 94

Also used in: Sheet: J-I 36; J-II 27, 33, 35, 41, 50, 55; J-M 11, 13, 17; C-II 20 | DPP: 5(6, 12-13) | Question Bank: 90, 92

Limits based on expansion

Question 55

What is the condition for existence of limit when numerator and denominator are expressed as expansion of x?

Model Answer

Finite limit $\iff$ lowest power of numerator $\geq$ lowest power of denominator; finite non-zero $\iff$ they are equal (limit = ratio of their coefficients).

Ex: $\dfrac{(\cos x-1)(\cos x-e^x)}{x^n}$: numerator $\approx\left(-\frac{x^2}{2}\right)(-x)=\frac{x^3}{2}$, so finite non-zero only for $n=3$.

Grade - Medium

Practice map, main concept: Sheet: J-I 36 | DPP: 5(22)

Also used in: Sheet: J-II 31, 33, 41; C-I 29 | DPP: 5(6) | Question Bank: 91

Question 56

What is the general instruction on variable for questions of limits based on expansion?

Model Answer

The expansions hold near 0: first make the variable tend to 0 ($x\to a$: $x=a+h$; $x\to\infty$: $x=\frac{1}{t}$), then expand each function up to the power of the denominator.

Ex: $\lim_{x\to\infty}\left(x-x^2\ln\left(1+\frac{1}{x}\right)\right)=\lim_{t\to 0}\dfrac{t-\ln(1+t)}{t^2}=\dfrac{1}{2}$.

Grade - Low

Practice map, main concept: none

Also used in: DPP: 5(14)

Question 57

What are the four standard questions on limits based on expansion and what is their answer?

Model Answer

$$\lim_{x\to 0}\frac{x-\sin x}{x^3}=\frac{1}{6},\qquad \lim_{x\to 0}\frac{\tan x-x}{x^3}=\frac{1}{3}$$

$$\lim_{x\to 0}\frac{e^x-1-x}{x^2}=\frac{1}{2},\qquad \lim_{x\to 0}\frac{\ln(1+x)-x}{x^2}=-\frac{1}{2}$$

Grade - High

Practice map, main concept: Sheet: J-I 32; J-II 27; J-M 14, 21 | Question Bank: 100, 102
Also used in: Sheet: J-II 37, 57; J-A 6, 8; C-I 18; C-II 11 | DPP: 5(4, 7, 11, 14-15)

Finding unknown constants

Question 58
How to find unknown constants in limit questions with the help of expansion?

Model Answer
Expand the numerator up to the power in the denominator; for a finite limit put the coefficients of all lower powers $= 0$, then the coefficient of the matching power gives the value.

Ex: $\lim_{x\to 0} \dfrac{ae^x - b\cos x + ce^{-x}}{x \sin x} = 2$: $a - b + c = 0$, $a - c = 0$, $\frac{a+b+c}{2} = 2 \Rightarrow a + b + c = 4$.

Grade - High
Practice map, main concept: Sheet: J-I 34; J-II 30-36, 41, 46, 50, 55; J-M 11, 13, 17; J-A 2-3, 5-6, 8; C-I 3, 8, 29; C-II 15 | DPP: 1(11); 2(12); 5(5-6, 8-9, 12-13) | Question Bank: 90-92, 96
Also used in: Sheet: J-II 42-43 | DPP: 2(13); 3(12); 5(3, 7)

Functions with built in limit

Question 59
How to define functions with built in limit?

Model Answer
Use $\lim_{n\to\infty} x^n = 0$ $(|x| < 1)$, 1 $(x = 1)$, ∞ $(x > 1)$, DNE $(x \le -1)$; split x into these cases and divide by the dominating term.

Ex: $f(x) = \lim_{n\to\infty} \dfrac{x^n}{x^n + 1}$ $(x > 0) = 0$ $(0 < x < 1)$, $\frac{1}{2}$ $(x = 1)$, 1 $(x > 1)$.

Grade - Medium
Practice map, main concept: Sheet: J-II 16; C-I 5; C-II 21, 23-25 | DPP: 5(18) | Question Bank: 108
Also used in: Sheet: J-II 22; J-M 1, 8; C-II 34 | Question Bank: 106

Miscellaneous

Question 60
When can we bring limit in the argument of a function in case of a composite function?

Model Answer
When the outer function f is continuous at $L = \lim g(x)$: $\lim f(g(x)) = f(\lim g(x))$. Works for ln, $e^{(\ )}$, $\sqrt{\ }$, polynomials, sin, ITF inside their domains; not for [ ], { }, sgn at a jump.

Ex: $\lim_{x\to 0} \ln \dfrac{\sin x}{x} = \ln 1 = 0$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-II 5, 54, 57; J-A 3; C-I 10 | DPP: 4(8); 5(12)

Question 61

What is the relation between logarithmic, polynomial and exponential function if x is approaching to infinity?

Model Answer

$\ln x \ll x^n \ll e^x$ $(n > 0)$: $\lim_{x\to\infty} \dfrac{\ln x}{x^n} = 0$, $\lim_{x\to\infty} \dfrac{x^n}{e^x} = 0$; hence also $\lim_{x\to 0^+} x \ln x = 0$.

Ex: $\lim_{x\to\infty} x^{10} e^{-x} = 0$.

Grade - Medium

Practice map, main concept: Sheet: J-I 25; J-A 13; C-I 10

Also used in: Sheet: J-II 52; C-I 7

Question 62

How to find a geometrical limit (length or area of a figure as an angle $\to 0$ or $n \to \infty$)?

Model Answer

Express each length/area in the parameter (sector $\frac{1}{2}r^2\theta$, triangle $\frac{1}{2}ab\sin C$, tangent length $r\tan\frac{\theta}{2}$), then use standard limits or expansions.

Ex: chord-arc area $A = \frac{r^2}{2}(\theta - \sin\theta)$, tangent-arc area $B = r^2\tan\frac{\theta}{2} - \frac{r^2\theta}{2}$: $\lim_{\theta\to 0} \dfrac{A}{B} = 2$.

Grade - Low

Practice map, main concept: Sheet: J-II 37; C-II 17-19, 26-28

Also used in: none